

- 7 (a) Half-life of a radioactive nuclide is defined as the time taken for half of the original number of radioactive nuclides in a sample to decay on average. **B1**

- (b) The beta decay is exothermic / releases energy. **B1**

Thus, the total binding energy of  ${}^{90}_{39}\text{Y}$  and  ${}^0_{-1}\text{e}$  is more than that of  ${}^{90}_{38}\text{Sr}$ .

**B1**

Since  ${}^0_{-1}\text{e}$  has no binding energy,  ${}^{90}_{39}\text{Y}$  has a greater binding energy. **B1**

- (c) (i) 1.  $A = \lambda N$

$$3.7 \times 10^6 = \frac{\ln 2}{(27.7)(365)(24)(60)(60)} N$$

**C1**

$$N = 4.66 \times 10^{15}$$

**A1**

2. 90 g of strontium contains  $6.02 \times 10^{23}$  atoms.

$$\text{Mass of } 4.66 \times 10^{15} \text{ atoms} = \frac{4.66 \times 10^{15}}{6.02 \times 10^{23}} (90/1000)$$

**C1**

$$= 6.97 \times 10^{-10} \text{ kg}$$

**A1**

**OR**

$$\text{Mass of } 4.66 \times 10^{15} \text{ atoms} = N(90u)$$

$$= (4.66 \times 10^{15}) (90) (1.66 \times 10^{-27})$$

**C1**

$$= 6.97 \times 10^{-10} \text{ kg}$$

**A1**

$$\text{Mass of } 4.66 \times 10^{15} \text{ atoms}$$

$$= N(38 \text{ mass of electrons} + 38 \text{ mass of protons} + 52 \text{ mass of neutrons})$$

$$= (4.66 \times 10^{15}) (38 \times 9.11 \times 10^{-31} + 38 \times 1.67 \times 10^{-27} + 52 \times 1.67 \times 10^{-27})$$

$$= 7.01 \times 10^{-10} \text{ kg}$$

- (ii)  $A = A_0 e^{-\lambda t}$

$$\frac{A}{A_0} = e^{-\left(\frac{\ln 2}{27.7}\right)(5.0)}$$

**C1**

$$= 0.882$$

**A1**

- 8 (a) Energy has discrete/fixed values

**B1**